Vo Nguyen Dang Khoa

CREATIVE MANAGER & AI INNOVATION LEAD

Visual Strategy • Generative AI Workflows • Art Direction • Full-Stack Engineering

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▣ SKILL MATRIX

◉ AI & GENERATIVE PRODUCTION

ComfyUI · FLUX.1 · Stable Diffusion · Midjourney v6 · Runway Gen-3 · LoRA Fine-Tuning · ControlNet · IP-Adapter · Prompt Engineering

◆ DESIGN, MOTION & 3D

Art Direction · Figma · Photoshop · After Effects · Premiere Pro · Cinema 4D · Blender · Design Systems · Visual Storytelling · Packaging

▲ LEADERSHIP & STRATEGY

Team Leadership (20+) · Omnichannel Visuals · eCommerce Growth · SOP Development · Brand Governance · Cross-functional Mgmt

⬢ ENGINEERING & CLOUD INFRA

React / Next.js · TypeScript · Go (Gin) · Rust · Docker · DevOps & CI/CD · Synology DSM · REST APIs · Linux

⌘ OPEN SOURCE & INFRA

V VietC IME
Modern Vietnamese input method engine for Linux with direct Unicode output and zero pre-edit lag.
Rust • Wayland • X11 • Linux

S Synology App Hub
Self-hosted private cloud ecosystem orchestrating micro-apps: kv-synology (DSM manager), kv-download, kv-netflix & kv-tube.
Next.js 15 • Go • Docker • HLS • SPK

C ComfyUI Node Suite
Custom ComfyUI nodes and automated workflows for e-commerce garment swap and multi-angle product rendering.
Python • PyTorch • CUDA • LoRA

⬢ EDUCATION & PROFILE

RMIT University Vietnam
Bachelor of Multimedia Design
Best Artistic Graduate Showcase (2016)
2012 – 2016 • English (Full Professional) • Vietnamese (Native)

Creative Manager & AI Innovation Lead | Visual Strategy & Creative Ops

Multidisciplinary Creative Lead with 9+ years transforming commercial video and brand production through strategic art direction and cutting-edge generative AI pipelines. Spearheaded enterprise design systems for Procter & Gamble across Southeast Asia and architected custom AI production workflows (ComfyUI, FLUX.1 LoRA, Runway Gen-3), slashing campaign turnaround by 60% without compromising commercial craft. Uniquely bridges senior creative direction, brand stewardship, and software engineering to deploy proprietary creative tools and automated production pipelines.

9+ Yrs	+60%	6 Markets	3× Faster
CREATIVE DIRECTION	OUTPUT ACCELERATION	REGIONAL REACH	TURNAROUND TIME
Campaign & Brand Strategy	Generative Video Workflows	P&G SEA Hair Care Portfolio	Rapid Concept to Launch

CREATIVE LEADERSHIP & WORK EXPERIENCE

9 YEARS • SEA REGION

AI Creative Manager — Phibious Vietnam
2025 – Present

— Direct commercial visual strategy and generative AI production for Fortune 500 brand campaigns, scaling asset throughput by 60% while maintaining uncompromising brand fidelity.

— Own the creative-tech roadmap and SOPs, architecting customized ComfyUI and FLUX.1 LoRA node pipelines that compressed campaign concept-to-pitch cycles from days to hours.

— Lead departmental P&L, production budgeting, and GPU infrastructure ROI; direct and mentor 20+ producers, art directors, and motion designers to maximize creative margins.

eCommerce Design Lead — Procter & Gamble (P&G)
2020 – 2025

— Directed end-to-end visual strategy and digital execution for P&G Hair Care portfolio (Head & Shoulders, Pantene, Rejoice) across 6 Southeast Asian regional markets.

— Engineered an enterprise modular design system with reusable UI/visual components, slashing quarterly asset adaptation time by 3× across 200+ campaign deliverables.

— Established automated QA review frameworks and partnered with growth marketing to optimize creative variants, achieving sustained double-digit CTR and conversion gains.

Production Creative Lead — INN SaiGon
2019 – 2020

— Led multimedia photography and video production for 30+ premier hospitality, lifestyle, and luxury F&B clients, reducing post-production turnaround by 40%.

— Oversaw quarterly production budgets up to $200K, establishing rigorous studio lighting, camera capture protocols, and high-end color grading standards.

Regional Head of Design — ASIAMARINE
2018 – 2019

— Orchestrated brand identity and luxury marketing collateral for Vietnam's premier yacht brokerage, elevating brand equity across regional clientele.

— Art directed the flagship 120+ page quarterly NAVIGATOR luxury publication, managing bilingual typography, bespoke editorial grid layouts, and press production.

FEATURED CREATIVE SYSTEMS & CAMPAIGNS

01 AI Commercial Video Pipeline
Automated generative production suite combining custom ComfyUI nodes, FLUX.1 LoRA adapters, and Runway Gen-3 for high-fidelity TVC & social video campaigns.
ComfyUI • FLUX.1 • LoRA • AnimateDiff • CUDA

02 P&G SEA Hair Care Design System
Scalable multi-market design system with 500+ master components deployed across Shopee, Lazada, and TikTok Shop for Head & Shoulders, Pantene, and Rejoice.
Art Direction • Design Systems • Figma • QA

03 Delux Luxury Perfume Campaign
End-to-end commercial campaign from AI moodboard generation to 3D bottle styling and final 4K video delivery, blending generative visuals with high-end motion.
Runway Gen-3 • Midjourney • LoRA • After Effects

04 NAVIGATOR Luxury Yacht Publication
Editorial layout, bespoke typography system, and interactive digital yacht catalog for ASIAMARINE's luxury clientele across Southeast Asia.
Editorial Direction • Typography • InDesign • Web

HONORS, RECOGNITION & CERTIFICATIONS

2024 **P&G SEA Digital Awards — Best Campaign**
2023 **Google Cloud — Professional Cloud Architect**
2016 **RMIT Best Artistic Graduate Showcase**

Vo Nguyen Dang Khoa

SECURITY CONSULTANT & SYSTEMS ARCHITECT

Information Security • Threat Modeling & DRM • Cryptographic Controls • Systems Engineering

syno.vndns.vn | github.com/vndangkhoa/vietc | github.com/vndangkhoa/kv-synology | vonguyendangkhoa@gmail.com | 0398300340 | Ho Chi Minh City, VN

▣ EXECUTIVE VALUE

Threat Modeling & Non-Normative Flows
Built forensic anti-leak DRM (ola) and kernel input isolation (vietc), designing runtime countermeasures against screen recording, DevTools inspection, and input race conditions.

Defense-in-Depth & Cryptography
Engineered Argon2id password hashing, RFC 6238 TOTP 2FA, Ed25519-signed offline licensing, and dunce::canonicalize filesystem sandboxing in kv-file-pro.

Compensating Controls & Supply Chain
Solved legacy GLIBC platform limitations via 100% static Musl cross-compilation in kv-trimui; enforced automated GnuPG package signing and dual-mirror CI/CD release assurance.

▣ TECHNICAL MATRIX

SECURITY & RISK
`Threat Modeling` `STRIDE` `Non-Normative Flows` `Forensic DRM` `Watermarking` `GRC` `Policy Enforcement`

CRYPTOGRAPHY & CONTROLS
`Argon2id` `RFC 6238 TOTP` `Ed25519` `GnuPG` `Sandboxing` `Rate Limiting` `CORS` `Zero-Trust`

SYSTEMS & LOW-LEVEL
`Rust 1.85+` `Linux (Arch/Debian)` `/dev/uinput` `Static Musl` `POSIX C` `Bash` `systemd rootless`

CLOUD & STORAGE
`Synology DSM 7.2+` `Docker Multi-Arch` `Go (Gin)` `React 19` `Next.js 15` `Btrfs RAID` `SQLite WAL`

▣ INFRASTRUCTURE ARCHITECTURE

Storage & Snapshot Architecture
Btrfs RAID storage pools with immutable snapshot replication, encrypted volume allocation, and automated retention lifecycle policies.

Gateway & Routing Infrastructure
Nginx reverse proxy with automated wildcard Let's Encrypt SSL certificate rotation, Hairpin-NAT, and Cloudflare DDNS routing.

▣ PRODUCTION ENDPOINTS

`https://syno.vndns.vn`
Synology Package Center Source

`github.com/vndangkhoa/vietc`
Rust Linux IME (60+ Stars, Wayland/X11)

`github.com/vndangkhoa/kv-synology`
Next.js 15 + React 19 DSM Control Plane

Security Consultant | Information Risk & Systems Architect

Security-minded systems and solutions engineer with 9+ years in technology and 4+ years designing security controls into production systems from the ground up: cryptographic licensing, forensic anti-leak DRM, multi-factor authentication, directory-traversal sandboxing, and tamper detection. Combines deep hands-on implementation (Go, Rust, TypeScript) with enterprise digital asset governance across Southeast Asia (Procter & Gamble). I approach every system as a consultant would: identify the threat model first, design controls proportional to risk, and document them so technical and non-technical stakeholders alike can make informed decisions.

7+ Apps
PRODUCTION SECURITY PACKAGES
Synology DSM 7.2+ Ecosystem

Ed25519
CRYPTOGRAPHIC CONTROLS
Argon2id + TOTP 2FA Sandboxing

Zero-Leak
FORENSIC ANTI-LEAK DRM
Dynamic Watermarking & Blanking

<10MB
MEMORY-SAFE RUST DAEMONS
Rootless /dev/uinput Subsystem

COMMUNITY STORE & CLOUD PLATFORM `SYNO.VNDNS.VN`

1-CLICK DSM APPS • SPK MONOREPO

Synology Community Store Distribution Platform

https://syno.vndns.vn

Production package repository (/mnt/data/Package Center/spk) serving 64-bit Synology DSM 7.2+ systems with automated cryptographic verification.

Infrastructure & Release Automation: Engineered an automated CI/CD release pipeline with GnuPG cryptographic verification to guarantee supply chain integrity. Features a zero-configuration deployment engine that automates container runtime validation, collision-free dynamic port allocation (20000–59999), and state/secret persistence across version upgrades—guaranteeing seamless updates with zero data loss.

kv-synology `FLAGSHIP DSM WEB MANAGER` — Next.js 15 • React 19 • TypeScript • Tailwind CSS v4

Architected a modern Synology DSM control plane featuring an intelligent **QuickConnect relay resolver** with automated fallback, real-time daemon controller for core protocols (SMB, NFS, SSH, WebDAV, AFP), live system notification streaming, volume health telemetry, and native 1-click DSM desktop launcher integration.

`Stack: Next.js 15 • React 19 • Tailwind CSS v4 • QuickConnect Protocol • Docker` — github.com/vndangkhoa/kv-synology

ola — Forensic Anti-Leak DRM
Dynamic moving watermark overlay (IP, timestamp, ID) and auto-blank traps upon screen-recording or DevTools inspection.
`Go • React • Synology SPK`

kv-file-pro — Cryptographic Sandboxing
Argon2id hashing, RFC 6238 TOTP 2FA, Ed25519 asymmetric offline licensing, and dunce::canonicalize path isolation.
`Rust (Axum) • SQLite WAL`

SysVis.AI — Threat Model Visualizer
Architecture topology and threat boundary mapping for Architecture Review Forums with local-first WebGPU AI inference.
`React 19 • WebLLM Qwen3`

kv-server — Zero-Trust Tunnel Gateway
Handheld reverse-proxy enforcing Cloudflare Zero-Trust Tunnels, token-based auth (?auth=token), and loopback port isolation.
`Rust • Musl • Cloudflared`

kv-tube — Hardened Media Engine
Per-IP rate limiting (300 req/min), strict CORS origin allow-lists, and multi-service container network segmentation.
`Go (Gin) • Next.js • HLS`

spkrepo — Supply Chain Packaging
Automated SPK package distribution platform with GnuPG cryptographic signature verification and container validation.
`GnuPG • Docker • Python`

SYSTEMS SOFTWARE • VIETC (VIET+) `RUST IME`

PURE RUST 1.85+ • WAYLAND & X11

Native High-Performance Linux Input Method Engine

github.com/vndangkhoa/vietc (60★)

Engineered from scratch in pure Rust to solve fundamental text composition and cursor flickering bugs across Wayland and X11.

Engineering Motivation: Traditional Linux IMEs relied on invasive pre-edit buffers that caused continuous cursor jumping, broke IDE autocompletion (VS Code, Cursor, Neovim), and suffered clipboard race conditions. Engineered a pure Rust solution leveraging Wayland's zwp_virtual_keyboard_v1 protocol and Linux /dev/uinput to inject native UTF-8 Unicode directly with sub-millisecond dispatch and zero clipboard interference.

Direct Wayland Virtual Keyboard
Direct Unicode keystroke injection via wtype (zwp_virtual_keyboard_v1) and /dev/uinput. Eliminates pre-edit underline flashing and preserves IDE autocompletion in VS Code and GPU terminals.
`wtype • /dev/uinput • <1ms Latency`

Intelligent Auto-Restore
Dual-layer Vietnamese phonological analysis paired with a technical English lexicon automatically preserves programming keywords (search, test, filter) without manual mode toggling.
`Phonology Engine • Context Awareness`

Hardware Event Filtering
Strict device path binding to /dev/input/by-path/*-event-kbd eliminates duplicate keystrokes caused by multi-interface 2.4GHz wireless USB dongles. Consumes <10MB RAM as a rootless daemon.
`Hardware Binding • <10MB Footprint`

SYSTEMS ENGINEERING & INFRASTRUCTURE MILESTONES

2024 — PRESENT

Secure Architecture & Forensic DRM (Ola & KV Files PRO) — 2024 — PRES
— Engineered forensic anti-leak DRM (ola) with dynamic moving watermarks (IP/time/ID) and auto-blank traps upon screen-recording or DevTools inspection.
— Built KV Files PRO with Argon2id password hashing, RFC 6238 TOTP 2FA, offline Ed25519 asymmetric licensing, and dunce::canonicalize filesystem sandboxing.

Memory Safety & Kernel Defense (VietC & KV-TrimUI) — 2024 — PRES
— Architected pure Rust 1.85+ input engine (vietc) with direct kernel /dev/uinput and /dev/input/by-path isolation, eliminating keystroke sniffing and duplicate injection (<10MB rootless).
— Engineered KV-TrimUI suite targeting static aarch64-unknown-linux-musl to overcome legacy GLIBC 2.28 vulnerabilities as a platform compensating control.

Regional Digital Asset Governance & Compliance (Procter & Gamble SEA) — 2020 — 2025
— Governed brand risk policies, asset validation, and regulatory compliance across 6 Southeast Asian regional markets (P&G Hair Care portfolio).
— Architected enterprise modular design system with 500+ master components, accelerating adaptation 3× while eliminating non-compliant asset deviations.

Vo Nguyen Dang Khoa

EMBEDDED SYSTEMS & HANDHELD OS ARCHITECT

Linux Kernel Subsystems • Bare Framebuffer Rendering • Rust Musl • Hardware Acceleration

trimui.vndns.net | github.com/vndangkhoa/kv-trimui | vonguyendangkhoa@gmail.com | 0398300340 | Ho Chi Minh City, VN

▣ EMBEDDED MATRIX

EMBEDDED LINUX & KERNEL

`Allwinner A133 (aarch64)` `Linux Framebuffer (/dev/fb0)` `Evdev Subsystem` `ALSA Direct PCM (hw:0,0)` `POSIX Signals` `Cedar HWDEC`

SYSTEMS & COMPILERS

`Rust 1.85+ (Musl)` `aarch64 Cross-Compile` `Static Binaries` `C / C99` `Bash IPC` `Double-Buffering`

HANDHELD RUNTIME

`Next.js 15 Static` `TinyHTTP` `Axum Daemons` `InnerTube API` `MPV Interop` `On-Screen Keyboard (OSK)`

DISTRIBUTION & NETWORK

`Wi-Fi OTA Updates` `MicroSD Packaging` `SQLite Manifests` `trimui.vndns.net` `REST Catalog API`

▣ SUBSYSTEM ARCHITECTURE

Bare-Metal Display & Input Loops
Bypassed missing display servers on stock Linux by memory-mapping /dev/fb0 with double-buffered BGRA rasterization, and handling low-level Linux Evdev controller inputs (/dev/input/event*) with repeat acceleration.

Musl Toolchains & Static Portability
Overcame legacy host GLIBC 2.28 barriers by engineering an automated cross-compilation pipeline targeting aarch64-unknown-linux-musl, deploying zero-dependency static binaries directly onto raw firmware.

Multimedia & Hardware Optimization
Integrated native MPV video pipes with GPU-assisted Cedar hardware decoding and direct ALSA PCM (hw:0,0) audio streaming, guaranteeing zero frame-drops and avoiding Android/stock OS audio resampling.

▣ PRODUCTION ENDPOINTS

`https://trimui.vndns.net`
Official OTA Handheld App Store Portal

`github.com/vndangkhoa/kv-trimui`
Complete Monorepo & Musl Daemons

`trimui.vndns.net/llms-full.txt`
Machine-Readable Framebuffer Specs

Embedded Systems Architect | Handheld Linux & Rust Engine Lead

Systems architect and low-level software engineer specializing in embedded Linux firmware, zero-overhead display pipelines, and resource-constrained aarch64 architectures. Creator of KV-TrimUI (github.com/vndangkhoa/kv-trimui)—a production-grade native operating ecosystem for Allwinner A133 handheld consoles (TrimUI Smart Pro / Brick). Engineered a dual-engine architecture combining a pure 100% Rust memory-mapped framebuffer engine (/dev/fb0, <900KB RAM) with static Musl micro-daemons (<15MB RAM), delivering 60 FPS hardware-accelerated video decoding (MPV), bit-perfect ALSA lossless audio, and an autonomous OTA software distribution network (trimui.vndns.net).

9 Apps
NATIVE HANDHELD SUITE
YouTube, Tidal, Netflix, AI, Radio

60 FPS
DIRECT FRAMEBUFFER BLIT
/dev/fb0 mmap • Zero X11/Wayland

< 1 MB
PURE RUST FOOTPRINT
kv-radio Embedded Memory Profile

720p 60f
HARDWARE ACCELERATED
Cedar Video Engine + Direct ALSA

HANDHELD CONSOLE SUITE & OTA ECOSYSTEM

`TRIMUI.VNDNS.NET` **9 PRODUCTION APPS • ALLWINNER A133**

KV-TrimUI Handheld Native Operating Ecosystem

`https://trimui.vndns.net`

Embedded aarch64 Linux software suite powering TrimUI Smart Pro and Brick consoles with zero-copy display output.

Engineering Invariants: Zero X11/Wayland dependencies. Direct memory-mapped framebuffer rendering (/dev/fb0, 1280×720 @ 60Hz 32bpp) with sub-millisecond input response via Linux Evdev. All native binaries statically compiled with Musl libc to eliminate GLIBC version mismatches across stock OS revisions.

kv-radio `PURE RUST 60 FPS FRAMEBUFFER`
`100% Rust • /dev/fb0 mmap • Direct ALSA • < 900 KB RAM`

Engineered a standalone embedded internet radio station player directly rendering to the Linux framebuffer. Implemented double-buffered page flipping, custom TrueType glyph rasterization, dirty-rectangle blitting, and streaming audio decode through direct ALSA PCM—consuming less than 900KB of system memory.

`Stack: Rust 1.85 • Linux Evdev • ALSA Direct • TrueType Rasterizer` `github.com/vndangkhoa/kv-trimui`

kv-launcher `PS5-STYLE SHELL`
60 FPS home launcher with real-time battery/Wi-Fi telemetry, ROM suggestions, and bilingual toggle.
`Next.js 15 • Rust Musl • Port 4540`

kv-tube `YOUTUBE CLIENT`
Ad-free streaming with InnerTube extraction, search auto-suggest, and 720p 60fps MPV acceleration.
`Next.js 15 • MPV HWDEC • YouTube.js`

kv-tidal `HI-FI AUDIOPHILE`
Bit-perfect lossless streamer supporting Tidal catalog, Apple Music Top 50, and Synology NAS Vault.
`Rust Daemon • Direct ALSA hw:0,0`

kv-netflix `HANDHELD CINEMA`
On-demand movie client with remote catalog scraping, 6-digit TV/phone pairing, and bilingual subtitles.
`HLS Streamer • Rust Musl • MPV`

kv-ai `VOICE AI COMPANION`
Gemini 2.5 & ChatGPT assistant with hardware Push-to-Talk voice, smartphone QR login, and D-Pad OSK.
`Push-to-Talk • Axum • WebSocket`

kv-file `PACKAGE INSTALLER`
Dual-pane handheld file manager, ZIP installer, and on-device client for the official OTA App Store.
`Next.js 15 • MicroSD Manager`

SYSTEMS & FIRMWARE MILESTONES

2026 ROADMAP

System Architect & Lead Engineer

`2026 — PRESENT`

KV-TrimUI Handheld Operating Platform

— Engineered the complete software suite for Allwinner A133 handhelds, replacing clunky emulators with a modern, ad-free connected media and tools suite.
— Built kv-radio in 100% pure Rust, implementing custom TrueType font rasterization, dirty-rectangle blitting, and streaming internet radio at <900KB RAM.
— Developed Push-to-Talk AI voice companion (kv-ai) with Gemini 2.5 & ChatGPT streaming, on-screen D-Pad virtual keyboard, and QR quick pairing.
— Architected the official OTA App Store & Distribution server (trimui.vndns.net) delivering one-tap Wi-Fi updates and package synchronization.

Core Engine Developer

`2026`

Unified Handheld Input & Multimedia Engine

— Created @kv-trimui/shared-gamepad layer mapping evdev joystick and button interrupts to standard gamepad events with sub-millisecond dispatch.
— Routed Tidal Hi-Fi and Apple Music streams directly through ALSA PCM without lossy downsampling, achieving true audiophile bit-perfect output.